

Yasmine

Senior Data Scientist & Machine Learning Engineer
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SUMMARY

Senior Data Scientist and Machine Learning Engineer with 5+ years of experience building, deploying, and optimizing production-grade AI/ML systems, Generative AI applications, LLM solutions, and scalable machine learning pipelines. Skilled in collaborating with data science and engineering teams to productionize models as secure, reliable, API-driven software products using Python, TensorFlow, PyTorch, Scikit-Learn, FastAPI, Flask, Docker, CI/CD, MLflow, AWS, Azure, and GCP. Experienced across the full ML lifecycle, including experimentation, model serving, workflow automation, monitoring, model optimization, RAG, LLMOps, MLOps, and enterprise AI deployment.

SKILLS

Machine Learning, Deep Learning, Generative AI, Large Language Models, Natural Language Processing, MLOps, LLMOps, Model Deployment, Model Serving, ML Pipelines, CI/CD Pipelines, Workflow Automation, Hyperparameter Tuning, Predictive Modeling, Statistical Analysis, Data Visualization, Python, Scikit-Learn, TensorFlow, Keras, PyTorch, Hugging Face Transformers, NLTK, OpenCV, Pandas, NumPy, PySpark, FastAPI, Flask, Streamlit, REST APIs, GraphQL, Docker, GitHub Actions, SQL, NoSQL, Pinecone, FAISS, Chroma DB, Neo4j, LangChain, LangGraph, LlamaIndex, RAG, AutoGen, LangSmith, AutoML, MLflow, AWS SageMaker, AWS Lambda, AWS Bedrock, AWS EC2, Azure ML, Azure AI, Azure App Services, GCP Vertex AI, GCP Cloud Functions, GCP BigQuery, Cloud Run, Agile Methodologies, Microservices Architecture, Stakeholder Management, Team Leadership

EXPERIENCE

Senior Data Scientist & Machine Learning Software Engineer - Generative AI - PURELOGICS

12/2024 - Present

- Designed, deployed, and optimized end-to-end AI/ML pipelines on AWS SageMaker, AWS Bedrock, AWS Lambda, GCP Vertex AI, BigQuery, and Cloud Run.
- Built enterprise-grade RAG pipelines, remote patient monitoring AI workflows, GPT-4o conversational AI systems, and agentic AI solutions for legal and financial advisory use cases.
- Productionized Generative AI and machine learning models into scalable, reliable, cloud-based software products with strong focus on availability, maintainability, and real-time insights.
- Engineered multi-agent AI architectures using LangChain, LangGraph, PhiData, Pinecone, and graph databases to automate decision-making workflows.
- Led cross-functional teams across AI strategy, LLMOps, MLOps, model lifecycle management, and production deployment.
- Mentored AI/ML teams on LLM fine-tuning, Retrieval-Augmented Generation, AI automation, and enterprise GenAI implementation.

Senior Data Scientist Machine Learning & Generative AI Engineer - Volunteer - NVIDIA

08/2024 - Present

- Contributed to GPU-optimized AI applications and enterprise Generative AI solution testing through NVIDIA and Z by HP developer programs.
- Developed and tested open-source and enterprise-level LLM solutions, including vLLM-based Generative AI applications.

- Integrated NVIDIA tools and SDKs to optimize AI workloads and validate scalable machine learning frameworks.
- Collaborated with technical teams to improve AI tooling, deployment workflows, and large-scale model performance.
- Supported open-source LLM initiatives focused on Generative AI innovation and large-scale deployment.

Senior Data Scientist & Machine Learning Engineer - BLOCBELT

12/2022 - 12/2024

- Led development of AI-driven SaaS and PaaS solutions while managing a team of three across big data, analytics, and AI engineering initiatives.
- Delivered AI and data engineering solutions that generated \$2M in client cost savings through workflow optimization and scalable automation.
- Improved sales forecasting accuracy by 80-90% through machine learning, statistical modeling, and analysis across 120+ market entities.
- Launched a personalized fashion e-commerce platform with an advanced recommendation engine, increasing sales by 80% and online orders by 200%.
- Developed predictive analytics models that contributed to an 18% revenue increase and improved data visibility by 45% through strategic dashboards.
- Integrated AI into SaaS and PaaS products, supported distributed systems, and executed 25+ A/B tests to improve digital customer experience.

AI Data Scientist Kaggle Master Mentor - Part Time - GOOGLE-KAGGLE

12/2022 - 11/2024

- Mentored aspiring data scientists through the Kaggle-X BIPOC Mentorship Program, dedicating 20 hours per week to technical coaching.
- Guided a team of four mentees in machine learning, data science workflows, and real-world problem-solving.
- Ranked among the Top 100 Kaggle contributors globally and contributed to high-impact AI and data science initiatives.
- Supported global community learning through applied ML projects, technical mentorship, and collaborative AI development.

Data Scientist - Crypto-Express

01/2022 - 12/2022

- Developed an anti-spoofing face application that enhanced digital identity security and reduced identity fraud by 70%.
- Applied machine learning and deep learning techniques to solve operational challenges, improving efficiency by 25% and reducing costs by 15%.
- Built predictive models and actionable analytics that supported data-driven decision-making and contributed to a 20% revenue increase.
- Performed EDA, machine learning modeling, and deep learning experimentation, saving 250+ hours of manual analysis annually.
- Deployed AI/ML projects remotely while adapting to dynamic requirements and production delivery needs.

Artificial Intelligence Engineer - Pakistan Freelancing Training Center

01/2021 - 12/2021

- Led AI and data science training sessions focused on practical machine learning implementation and real-world problem-solving.
- Designed and deployed models using TensorFlow and Keras to improve system efficiency and prediction accuracy.
- Automated repetitive data tasks using Python, reducing manual effort and minimizing error rates.
- Supported applied AI research and translated emerging methodologies into practical project implementations.

PROJECTS

AutoML-Studio

Built a low-code/no-code machine learning platform for rapid predictive model development and deployment. Integrated MLflow for experiment tracking, SHAP for explainability, data drift detection for model reliability, and FastAPI and Streamlit for production-ready serving across classification, regression, time-series forecasting, and clustering.

Customer Satisfaction Prediction System

Created a predictive analytics system using ZenML and MLflow within an MLOps framework, achieving 93% accuracy in forecasting customer satisfaction and streamlining pipeline management for business decision-making.

Conversational AI Chatbot for Customer Support

Designed and deployed an enterprise-grade chatbot powered by GPT-4o, LangChain, and AWS Bedrock. Implemented intent recognition and dynamic response generation, reducing customer query resolution time by 40%.

Health Assistance Application

Architected a health assistance application using Gemini 1.5 Pro, RAG, and LangChain to support doctors and patients with medical information, symptom diagnosis, treatment suggestions, and preventive healthcare advice.

DocuWiz AI

Engineered a document intelligence tool using BART, DistilBERT, RAG, and LLMs for text extraction, summarization, sentiment analysis, and PDF/DOCX insights. Deployed with Streamlit and FastAPI, reducing document processing time by 60%.

Intelligent Document Summarization Tool

Developed a transformer-based document summarization system using Pegasus, FastAPI, and Streamlit to generate concise summaries from lengthy PDF and DOCX documents, reducing manual review time by 60%.

AI-Powered Blog Generator

Developed an AI content generation platform using Llama 3.1 8B, AWS Bedrock, RAG, and Streamlit to automate blog creation, editing, and publishing for enterprise content workflows.

Harvestify AI-Powered Plant Health Assistant

Designed an AI application using Azure AI Studio and Azure App Services to diagnose plant leaf diseases through real-time image analysis, recommend crops and fertilizers, reduce disease detection time by 70%, and improve crop yield predictions by 30%.

EDUCATION

Bachelor in Computer Science, Major: Artificial Intelligence and Computer Science - Institute of Management Sciences